

Practice Projects

For practice, write a program for each of the following tasks:

1. Create a turtle program that draws a box fractal as shown in [Figure 9-20](#). This program is similar to the Sierpiński carpet program introduced in this chapter. Use the `turtle.begin_fill()` and `turtle.end_fill()` functions to draw the first large, black square. Then split this square into nine equal sections, and draw white squares in the top, left, right, and bottom squares. Repeat this process for the four corner squares and the center square.

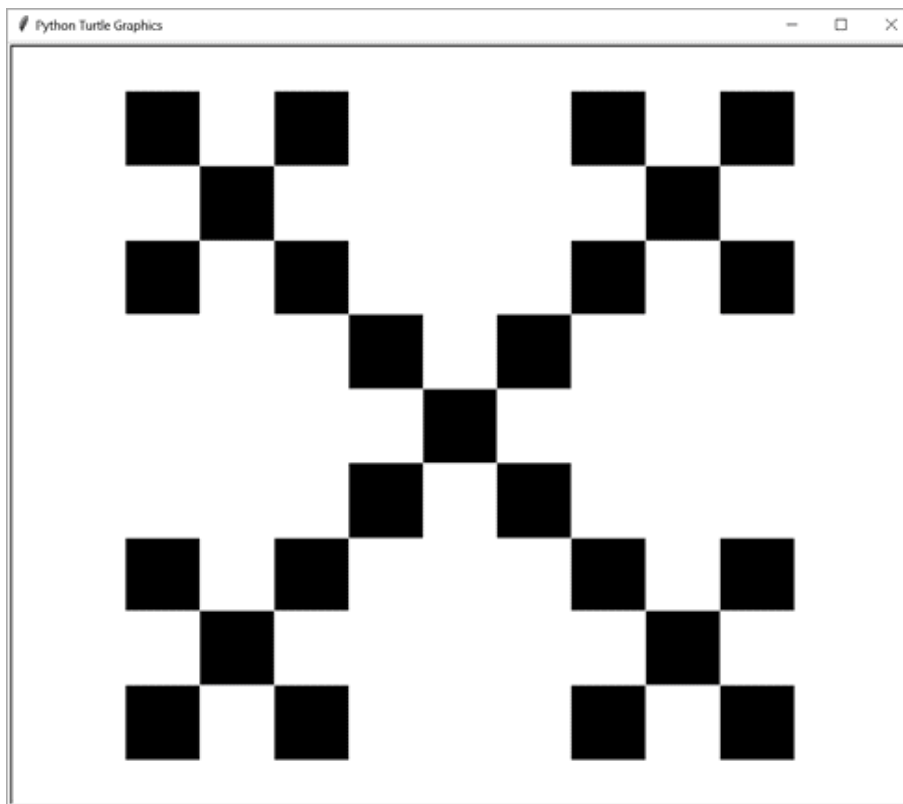


Figure 9-20: A box fractal, drawn to two levels

2. Create a turtle program that draws a Peano space-filling curve. This is similar to the Hilbert curve program in this chapter. [Figure 9-21](#) shows the first three iterations of the Peano curve. While each Hilbert curve iteration is split across a 2×2 section (which is, in turn, split into 2×2 sections), the Peano curve is split across 3×3 sections.